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MAJOR PROJECT REGISTRATION FORM

Date:

Project Category:	☐ Research oriented ☐ Application oriented ☐ Product oriented ☐ Other (please specify) Specify if others:	Batch ID:	AIML-B17

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Title of the Project: Securing Direct-to-Cloud File Uploads in Modern Web Applications Using Autonomous AI Agents

Abstract

With the increasing adoption of cloud storage for handling large volumes of user files, modern web applications increasingly use **direct-to-cloud file uploads**, where users upload files directly from their browsers to cloud storage services without routing the files through the application server. Although this approach reduces server-side workload, network overhead, and improves upload efficiency and scalability, it also introduces security risks involving authentication, upload credentials, credential validity, file type and size restrictions, access control, file overwriting, and upload callbacks. Traditional security assessment of these workflows often relies on manual inspection and predefined security tests, making the process time-consuming and difficult to scale. To address these challenges, we propose an **autonomous AI-agent-based security assessment framework** that operates from the **client/browser side** through a browser-based security extension. The framework employs two specialized agents: a **Security Planner Agent**, which observes the upload workflow, analyzes the available upload parameters, and plans appropriate security tests, and a **Security Verification Agent**, which evaluates the collected responses and evidence to verify whether a suspected weakness represents an actual security vulnerability. Controlled security-testing tools execute the planned tests, while the collected evidence is analyzed to assess authentication, credential validity, file restrictions, access permissions, overwrite behavior, and callback security. By combining autonomous AI agents, browser-based integration, and automated security testing, the proposed framework aims to reduce manual effort, improve the consistency and scalability of security assessment, and generate structured evidence-based security reports. The framework is evaluated in a controlled environment to demonstrate its capability to automatically identify and verify security weaknesses in modern web applications that use direct-to-cloud file upload mechanisms.

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